

Problem statement- assignment 2

1. Where would you rate yourself on (LLM, Deep Learning, AI, ML). A, B, C [A = can code independently; B = can code under supervision; C = have little or no understanding]

- B = can code under supervision. I know basics about LLM, Deep Learning, NLP, Memory Management & its types, AI, ML, RAG etc. I also developed one demo application of HR Policy Chat Bot which answers employees' questions regarding company's policy from pre-defined set of policy by the company.

2. What are the key architectural components to create a chatbot based on LLM? Please explain the approach on a high-level

- User -> Chatbot/API -> Context/Knowledge -> LLM -> Response -> User
- At a high level, an LLM chatbot has a UI, a backend API, conversation memory, an optional RAG layer for company-specific knowledge, and an LLM.

When a user sends a question, the backend receives it and adds the relevant conversation history. If the chatbot needs information from our internal documents, we retrieve the relevant content from a vector database. We then send the user's question, conversation context, and retrieved information to the LLM.

If the LLM needs to perform an action, such as querying an order database or calling an external API, we provide it with tools that it can invoke. Finally, the LLM generates the response, which goes back through the backend to the user.

Around all of this, we have security, monitoring, logging, and rate limiting to think of.

3. Please explain vector databases. If you were to select a vector database for a hypothetical problem (you may define the problem) which one will you choose, and why?

- A Vector Database stores data as numerical format so we can find information based on meaning of the words or sentences, not just exact keywords. For example, for a company chatbot that answers questions from internal documents, I would choose Pinecone because it is easy to scale and provides fast semantic search. The documents are converted into embeddings and stored in Pinecone, and when a user asks a question, we find the most relevant documents using vector similarity. I would choose it when we expect the knowledge base and user traffic to grow, while for a small/local project I might choose FAISS or Chroma instead.